Table 4: Hyperparameter tuning results for Algebra 2005–2006

Corrected causal LightSAKT · validation results

Learning Rate	Embedding	Heads	Dropout	Best validation AUC
0.000500	32	2	0.1	0.788834
0.000500	32	2	0.2	0.787610
0.000500	32	4	0.1	0.789701
0.000500	32	4	0.2	0.788728
0.000500	64	2	0.1	0.800535
0.000500	64	2	0.2	0.799574
0.000500	64	4	0.1	0.801017
0.000500	64	4	0.2	0.800187
0.000500	128	2	0.1	0.807181
0.000500	128	2	0.2	0.807049
0.000500	128	4	0.1	0.807573
0.000500	128	4	0.2	0.807942
0.001000	32	2	0.1	0.797397
0.001000	32	2	0.2	0.796752
0.001000	32	4	0.1	0.797403
0.001000	32	4	0.2	0.796880
0.001000	64	2	0.1	0.803621
0.001000	64	2	0.2	0.803011
0.001000	64	4	0.1	0.804018
0.001000	64	4	0.2	0.803866
0.001000	128	2	0.1	0.807105
0.001000	128	2	0.2	0.807842
0.001000	128	4	0.1	0.807769
0.001000	**128**	**4**	**0.2**	**0.808127**
0.005000	32	2	0.1	0.801522
0.005000	32	2	0.2	0.801106
0.005000	32	4	0.1	0.801053
0.005000	32	4	0.2	0.800365
0.005000	64	2	0.1	0.803593
0.005000	64	2	0.2	0.802481
0.005000	64	4	0.1	0.804632
0.005000	64	4	0.2	0.803808
0.005000	128	2	0.1	0.807475
0.005000	128	2	0.2	0.807133
0.005000	128	4	0.1	0.807367
0.005000	128	4	0.2	0.807673

LightSAKT; all 36 configurations. Training seed 42, split seed 42; validation AUC only.
Bold marks the highest full-precision validation AUC. Exact ties select the lexically smallest run ID.
Selected: LR 0.001, dimension 128, heads 4, dropout 0.2; epoch 26.